Gabriel MARIADASS | Paris, France

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SUMMARY

M.Sc. MVA at ENS Paris-Saclay and M.Eng. at École Nationale des Ponts et Chaussées, with applied ML experience at Dassault Systèmes and Banque de France and research in uncertainty-aware ML at LPSM Paris and NTU Singapore. Seeking an ML/AI Engineering role in Paris, and open to research-oriented positions such as a CIFRE PhD. Available immediately.

EDUCATION

ENS Paris-Saclay, France *2025 - 2026*
M.Sc. Mathematics, Vision, and Machine Learning (MVA)

- **Academic average**: 17.19/20 (with Highest Honors)
- **Key courses**: Deep Learning, ALTEGRAD (Text and Graph Learning), Graphs in ML, Time Series ML, Bayesian ML, Computational Statistics, Convex Optimization for ML, Learning with Extreme Values

École Nationale des Ponts et Chaussées, France *2022 - 2026*
M.Eng. - Mathematical and Computer Engineering (IMI)
Final year completed at ENS Paris-Saclay (MVA)

Lycée Henri-IV, France *2020 - 2022*
*CPGE - PCSI/PC**

PROFESSIONAL EXPERIENCE

Dassault Systèmes, France *01/2025 - 07/2025*
ML Engineer - GNN for Molecular Retrosynthesis

- Migrated a GNN retrosynthesis pipeline from DGL to PyG, up to 38% faster inference at comparable top-1 (66.5% -> 66.1%)
- Evaluated Bayesian hyperparameter optimization (+1.5 points top-1) and rejected it (2.8x larger model)
- Tested dynamic and static quantization and refactored the codebase into a single configurable pipeline

Banque de France, France *09/2024 - 12/2024*
Data Scientist - Credit Risk

- Built neural-network challenger models on 700K loans, reaching AUC 0.84 and +0.14 Gini over the bank's internal IRB models
- Analyzed default-rate time series (PCA, dynamic factor model) and PD margins of conservatism for a live ECB-mandated audit

RESEARCH EXPERIENCE

LPSM, France & NTU, Singapore *04/2026 - 08/2026*
ML Research Intern - Possibilistic Variational Inference
Supervisors: Badr-Eddine Chérief-Abdellatif (CNRS, LPSM) and Jérémie Houssineau (NTU)

- Reproduced and extended uCBOpt (uncertainty-aware optimizer) with learned precision, benchmarked vs IVON and AdamW
- Improved NLL and ECE (0.025 -> 0.014-0.022) over uCBOpt, and beat 64-sample IVON on ECE (0.014 vs 0.019) with a single pass
- Extended an existing SAM connection with an adaptive, learned perturbation radius
- Documented the numerical limits of learned precision at network scale

TECHNICAL SKILLS

- **Programming**: Python, SQL, Bash
- **ML Engineering** (see price-api): Docker, FastAPI, CI/CD (GitHub Actions), Cloud (GCP), MLflow, pytest, Git
- **ML & Deep Learning**: PyTorch, scikit-learn, XGBoost/LightGBM, NumPy, Pandas, tabular modeling, time-series forecasting, Transformers/LLMs, RAG, LoRA fine-tuning, GNNs (PyG, DGL), calibration/conformal prediction, Optuna, quantization
- **Mathematics**: Probability & Statistics, Bayesian Learning, Convex Optimization, Possibility Theory, Extreme Values

LANGUAGES & INTERESTS

- **Languages**: French (native), English (C1), Spanish (B1)
- **Interests**: Muay Thai, Running, Weightlifting, Nutrition, AI Tech Watch, Tech Entrepreneurship, Reading, Travel